

Day 9: Star Schema & Dimensional Modeling

[DEF] Definition to Remember

A Star Schema is the gold-standard dimensional data modeling architecture for Power BI where a central Fact Table (containing numeric measures and keys) is surrounded by multiple Dimension Tables (containing descriptive business context).

1. Fact Tables vs Dimension Tables

Feature	Fact Table	Dimension Table
Content	Quantitative metrics, transactions, nu	Descriptive text attributes, categorie
Examples	`Fact_Sales`, `Fact_Orders`, `Fact_Inv	`Dim_Customer`, `Dim_Product`, `Dim_Da
Granularity	Transactional event level (e.g., one r	Entity level (e.g., one row per unique
Row Count	Millions / Billions of rows (Tall & Na	Hundreds / Thousands of rows (Short &

2. Star Schema vs Snowflake Schema

STAR SCHEMA (Recommended for Power BI):

```

    [Dim_Product]   [Dim_Customer]
      \             /
       [Fact_Sales]
      /             \
[Dim_Date]         [Dim_Store]
```

SNOWFLAKE SCHEMA (Normalized dimensions -- sub-tables):

```
[Dim_Category] -> [Dim_SubCat] -> [Dim_Product] -> [Fact_Sales]
```

3. The Dedicated Date Dimension Table (`Dim\_Date`)

- * Mandatory for all Power BI models to support accurate DAX Time Intelligence.

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[MUST-WRITE] Must-Master Skills & Interview Points

1. Why Power BI's VertiPaq engine is heavily optimized for Star Schema architectures.
2. Why Snowflake schemas should be flattened into a Star Schema in Power Query.
3. Why every professional BI model must have a dedicated, marked Date table.

[RECALL] Quick Recall

Central Fact Table + Surrounding Dimension Tables + Dedicated Date Dimension = Peak Power BI Performance